

AL-2002 - Artificial Intelligence Lab

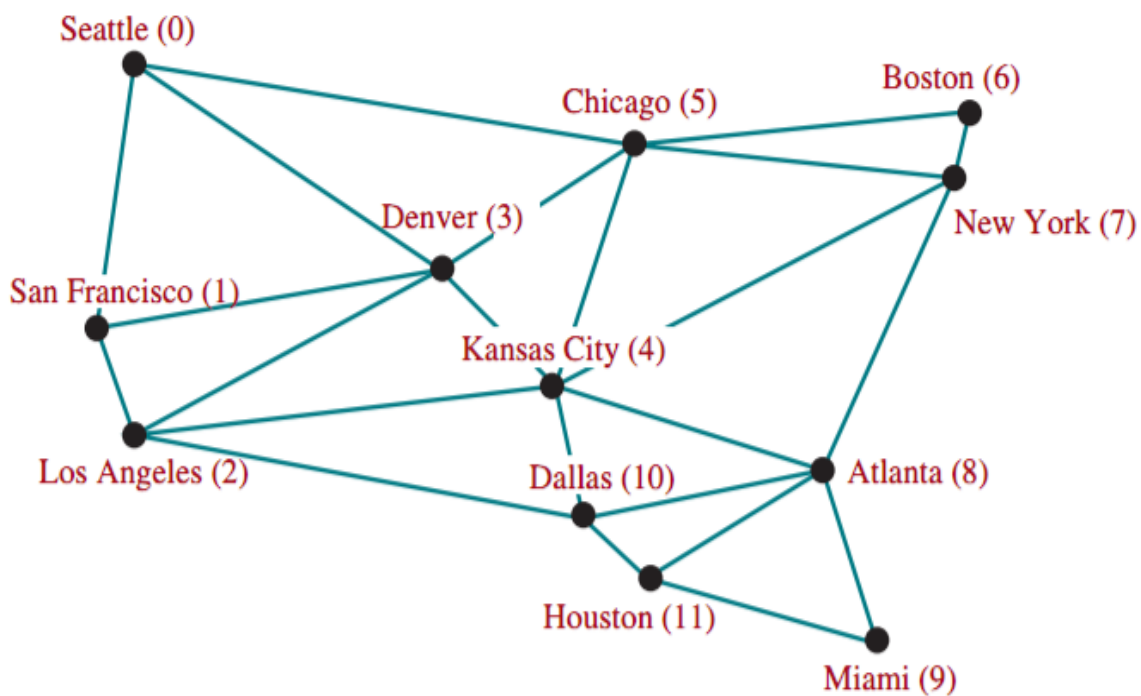
Lab # 2

Objectives:

- State Space Representation + Uninformed Search Algorithms (BFS+DFS+DLS+IDDFS)

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in multiple cells of the same notebook.
3. **Must Use Functions and Exception handling for each question, where they can be applied.**
4. **Do not copy from any source otherwise you will be penalized with negative marks.**
5. Add your source code in word document and attach output screenshots.
6. Upload word file and ipynb file.
7. Please submit your **Both files** with this naming convention ROLLNO_SECTION_LABNO. (**Do not upload zip file**).
8. Submit your lab on Google Classroom.



Question 1:

A flight booking system needs to determine the minimum number of flights required to travel

between two cities. Each city is represented as a node in a graph, and each direct flight between two cities is represented as an edge. Passengers prefer routes with the fewest number of connecting flights. Given a graph representing cities and their direct flight connections, write a Python program to implement Breadth First Search (BFS) to find the shortest path (in terms of number of flights) from a given source city to a destination city. Your program should display the path taken and the total number of flights required.

Question 2:

In a disaster response scenario, emergency services need to explore all possible evacuation routes from a starting city to reach a safe location. Cities are represented as nodes, and roads between cities are represented as edges in a graph. Due to limited initial information, the system explores one route completely before trying alternative routes. Write a Python program to implement Depth First Search (DFS) to determine a path from a given starting city to a goal city. The program should print the path discovered by DFS and demonstrate how the algorithm explores deeper paths before backtracking.

Question 3:

A submarine rescue mission needs to locate a trapped vessel within a limited search depth. The rescue team can only deploy resources up to a certain number of hops away from base at a time, gradually increasing their search radius if the vessel isn't found. Cities represent underwater checkpoints and edges represent navigable routes between them.

Write a Python program to implement **Iterative Deepening Depth First Search (IDDFS)** to find a path from the rescue base to the trapped vessel. The program should display each depth limit being attempted, the nodes explored at that depth, and the final path when found.